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computer vision-ITA0501

Short Answer Test

1) Compare two image acquisition systems (CCD vs CMOS) in terms of their impact on image quality in practival applications.

CCD provides better image quality with low noise but is expensive. CMOS is faster, cheaper consumes less power.

2) ~~Adi~~ Causes of image distortion and improvements

Possible causes:- Poor illumination, lens distortion, sensor noise

Improvements:- use proper lighting, Image filtering, lens correction

3) Computer vision features in automated surveillance

features:- object detection, face recognition, Motion detection.

Applications:- Detect intruders, Track suspicious pessons, Alests

4) Satellite image processing accuracy

Digital image fundamentals:-

- Highes spatial resolution captures more detailes.

→ Better spectral resolution Land types.

Image acquistion methods:- High quality sensors, correct calibration, suitable viewing angle.

5) OCR: effect of sampling, quantization and image acquisition.

Sampling:- Higher sampling captures clear character edges.

Quantization:- More gray levels preserve character details.

Image acquisition:- proper lighting and focus reduce noise.

6) Reduce storage while maintaining quality.

→ slightly reduce image resolution.

→ Apply image compression.

Benefits:- Lower storage, ~~faster~~ processing.

7) Uneven lighting and poor contrast.

cause:- Non-uniform illumination during image formation.

Corrective measures:-

→ Improve lighting conditions.

→ Apply histogram equalization

8) Improper quantization in medical imaging

Effects:- Losse of subtle tissue details, Reduced contrast.

Solution:-

- Use higher bit-depth
- preserve intensity information.

9) Self-driving cars:- Image acquisition and computer vision

Image acquisition:-

1. High resolution cameras.
2. fast frame rate

Computer vision:-

→ Lane detection, Traffic sign recognition, obstacle detection

10) Detecting fine image details

Sampling rate:- use a high sampling rate to capture fine structures

Intensity resolution:- use higher bit depth to distinguish subtle brightness differences.